



BIS Standards Recommendation Engine

AI-Powered Compliance Discovery for Indian Micro & Small Enterprises

From product description to BIS standard — in seconds.

Team name:2HACK

TEAM MEMBERS

Abhigyan Singh | Waseem Raza

Helping Indian MSEs find the right BIS standard instantly — using RAG, BM25 + FAISS hybrid retrieval.

BIS × SS Hackathon • May 2026 • Track: AI / RAG

01 PROBLEM STATEMENT

MSEs Spend Weeks on What Should Take Seconds

63M+

MSMEs in India

Only ~5% have dedicated
compliance teams

3-6

Weeks Lost

Average time to identify
applicable BIS standards

26,000+

BIS Standards

Across categories — nearly
impossible to search manually

For Building Materials (Cement, Steel, Concrete, Aggregates), non-compliance means rejected shipments, factory shutdowns, and loss of BIS certification — crippling for small enterprises with limited resources.

The gap: No affordable, fast, AI-powered tool exists for MSEs to self-serve BIS discovery.

One Query. Instant BIS Standards. Full Rationale.



Accept

A natural language product description
(e.g. 'high strength cement for construction')



Retrieve

Top-K semantically similar BIS standard
chunks from a FAISS vector index



Rerank

Hybrid scoring: semantic similarity
+ keyword overlap for precision



Explain

LLM generates a 1–2 sentence rationale
per standard — grounded in the text

Sample Output

#1 — IS 8112 : 1989

This standard specifies requirements for 43 grade ordinary Portland cement used in general construction, matching your query for high-strength cement applications.

#2 — IS 1489 : Part 1 : 1991

Covers Portland pozzolana cement (fly ash based) with strength properties relevant to structural concrete applications.

#3 — IS 455 : 1989

Portland slag cement standard applicable for construction requiring moderate heat of hydration and sulfate resistance.

Corrected End-to-End RAG Pipeline



Tech Stack

Embedding	SentenceTransformers all-MiniLM-L6-v2
Vector Store	FAISS IndexFlatIP (cosine / exact search)
Keyword	BM25 (custom impl — no external lib)
Fusion	0.4 × BM25 + 0.6 × Dense score merge
PDF Parse	PyMuPDF (fitz) — paragraph-aware
LLM (app)	Groq Llama 3.1 8B — app.py only

Data Ingestion (One-Time) + Corrections

- BIS SP 21 PDF → PyMuPDF Parser
- Standard-boundary chunker (~568 chunks)
- MiniLM Encoder → FAISS IndexFlatIP
- BM25 built at runtime from metadata.pkl

Why Our Approach Works



Chunking Strategy

- ✓ Standard-boundary chunking — one chunk $\approx$ one IS standard summary
- ✓ PyMuPDF (fitz) preserves paragraph structure vs naive pypdf splits
- ✓ Each chunk retains the IS XXXX identifier for regex extraction
- ✓ Avg chunk size: ~300–500 tokens — fits LLM context cleanly
- ✓ No overlap needed: standard summaries are self-contained units



Hybrid Retrieval

Stage 1 — Semantic Search

FAISS IndexFlatL2 with MiniLM embeddings.
Searches k=10 candidates via cosine similarity.

Stage 2 — Keyword Reranking

Scores each candidate by counting query-word overlaps in chunk text. Promotes exact matches.

Stage 3 — Deduplication

Filters repeated IS numbers. Guarantees distinct standards in the final top-5 output.

Stage 4 — LLM Rationale

Groq Llama 3.1 8B explains relevance using only the chunk text — no hallucination risk.

Live Queries on BIS SP 21 Dataset

Query 1: "high strength cement for building construction"

 1.8s

#1 IS 8112 : 1989 — 43 Grade OPC

#2 IS 1489 : Part 1 — PPC Fly Ash

#3 IS 455 — Portland Slag Cement

Query 2: "reinforcement steel bars for RCC slabs"

 2.1s

#1 IS 1786 : 2008 — HYSD Steel Bars

#2 IS 432 : Part 1 — Mild Steel Bars

#3 IS 2502 — Steel Bar Bending

Query 3: "coarse aggregate for concrete mix design"

 2.3s

#1 IS 383 : 2016 — Coarse Aggregates

#2 IS 2386 : Part 1 — Aggregate Testing

#3 IS 10262 — Concrete Mix Design

06 EVALUATION RESULTS

Automated Metrics — Public Test Set (10 Queries)

Hit Rate @3

100%

Target: >80%

✓ Exceeds Target

MRR @5

0.85

Target: >0.70

✓ Exceeds Target

Avg Latency

0.08s

Target: <5 sec

✓ Well Within Target

Manual Evaluation (Semi-Automated):

Relevance Score

4.2 / 5.0

Random 5-query judge review

Hallucination Rate

0%

No imaginary standards detected

Real Impact for Indian MSEs



Weeks → Seconds

Standard discovery time reduced from 3–6 weeks to under 5 seconds



Zero Cost

Built on open-source models + free-tier Groq API — no licensing fees



Self-Service

MSEs can query without compliance experts or government consultants



Grounded Answers

LLM constrained to document text — no hallucinated standards



Our Team



Waseem Raza

UI/UX, Testing & Documentation



Abhigyan Singh

RAG Pipeline, Evaluation & Data Ingestion



Acknowledgements

Bureau of Indian Standards (BIS)
Hugging Face — sentence-transformers
Groq — LLM inference platform

Future Roadmap:

Multi-category expansion • Regional language support • BIS portal integration • Mobile app